

Section B: API Endpoint Plan & Database Architecture

Complete Endpoint Specifications, Database Schema Script & Seed Data

1. Complete API Endpoint Plan

METHOD	ROUTE	DESCRIPTION	ROLE REQUIRED	REQUEST BODY	EXPECTED RESPONSE
POST	/api/auth/register	Register a new user (Organiser or Participant)	None (Public)	{ "email", "password", "fullName", "role", "organisationName" (if organiser), "dateOfBirth" (if participant) }	201 Created - { "userId", "email", "role" } 400 Bad Request - Invalid data 409 Conflict - Email already exists
POST	/api/auth/login	Authenticate user and create session	None (Public)	{ "email", "password" }	200 OK - { "userId", "email", "role" } 401 Unauthorized - Invalid credentials
POST	/api/auth/logout	End user session	Any (Logged in)	None	200 OK - { "message": "Logged out successfully" }
GET	/api/users/profile	Get current user's profile	Any (Logged in)	None	200 OK - User profile object 401 Unauthorized - Not logged in
PUT	/api/users/profile	Update current user's profile	Any (Logged in)	{ "fullName", "email", "organisationName" (if organiser), "dateOfBirth" (if participant) }	200 OK - Updated user profile 400 Bad Request - Invalid data
GET	/api/events	Get all events with optional filters	None (Public)	None (Query params: type, date, location)	200 OK - List of events
GET	/api/events/{id}	Get specific event details	None (Public)	None	200 OK - Event details with categories 404 Not Found - Event doesn't exist
POST	/api/events	Create a new event	Organiser	{ "name", "description", "date", "location", "distance", "eventType", "bannerImageUrl" }	201 Created - New event object 400 Bad Request - Invalid data
PUT	/api/events/{id}	Update an existing event	Organiser	{ "name", "description", "date", "location", "distance", "eventType", "bannerImageUrl" }	200 OK - Updated event 403 Forbidden - Not the event organiser 404 Not Found - Event doesn't exist
DELETE	/api/events/{id}	Delete an event	Organiser	None	204 No Content 403 Forbidden - Not the event organiser 404 Not Found - Event doesn't exist
GET	/api/events/{eventId}/categories	Get all categories for an event	None (Public)	None	200 OK - List of categories 404 Not Found - Event doesn't exist

METHOD	ROUTE	DESCRIPTION	ROLE REQUIRED	REQUEST BODY	EXPECTED RESPONSE
POST	/api/events/{eventId}/categories	Create a new category for an event	Organiser	{ "name", "description", "minAge", "maxAge" }	201 Created - New category object 400 Bad Request - Invalid data 404 Not Found - Event doesn't exist
PUT	/api/categories/{id}	Update a category	Organiser	{ "name", "description", "minAge", "maxAge" }	200 OK - Updated category 403 Forbidden - Not the event organiser 404 Not Found - Category doesn't exist
DELETE	/api/categories/{id}	Delete a category	Organiser	None	204 No Content 403 Forbidden - Not the event organiser 404 Not Found - Category doesn't exist
GET	/api/enrolments	Get all enrolments (filtered by role)	Any (Logged in)	None	200 OK - List of enrolments (organisers see all, participants see own)
GET	/api/events/{eventId}/enrolments	Get all enrolments for a specific event	Organiser	None	200 OK - List of enrolments 404 Not Found - Event doesn't exist
POST	/api/events/{eventId}/enrol	Enrol a participant in an event	Participant	{ "categoryId" }	201 Created - Enrolment object 400 Bad Request - Already enrolled or category invalid 404 Not Found - Event doesn't exist
PUT	/api/enrolments/{id}/status	Update enrolment status	Organiser	{ "status" }	200 OK - Updated enrolment 403 Forbidden - Not the event organiser 404 Not Found - Enrolment doesn't exist
GET	/api/results/me	Get current participant's results	Participant	None	200 OK - List of participant's results
GET	/api/events/{eventId}/results	Get all results for an event	Organiser	None	200 OK - List of results 404 Not Found - Event doesn't exist
POST	/api/events/{eventId}/results	Capture results for a participant	Organiser	{ "enrolmentId", "finishTime", "finishPosition" }	201 Created - Result object 400 Bad Request - Invalid data 404 Not Found - Enrolment doesn't exist
PUT	/api/results/{id}	Update a result	Organiser	{ "finishTime", "finishPosition" }	200 OK - Updated result 403 Forbidden - Not the event organiser 404 Not Found - Result doesn't exist

METHOD	ROUTE	DESCRIPTION	ROLE REQUIRED	REQUEST BODY	EXPECTED RESPONSE
POST	/api/events/{eventId}/upload-banner	Upload event banner image	Organiser	FormData with image file	200 OK - { "imageUrl" } 400 Bad Request - Invalid file

2. Database Schema Creation (SQL Script)

```
-- Create Database
CREATE DATABASE RaceDayDB;
GO
USE RaceDayDB;
GO

-- 1. Users Table
CREATE TABLE Users (
    UserID INT IDENTITY(1,1) PRIMARY KEY,
    Email NVARCHAR(255) UNIQUE NOT NULL,
    PasswordHash NVARCHAR(255) NOT NULL,
    FullName NVARCHAR(100) NOT NULL,
    Role NVARCHAR(20) NOT NULL CHECK (Role IN ('Organiser', 'Participant')),
    CreatedAt DATETIME DEFAULT GETDATE()
);
GO

-- 2. Organisers Table
CREATE TABLE Organisers (
    OrganiserID INT IDENTITY(1,1) PRIMARY KEY,
    UserID INT UNIQUE NOT NULL,
    OrganisationName NVARCHAR(100) NOT NULL,
    ContactNumber NVARCHAR(20) NOT NULL,
    FOREIGN KEY (UserID) REFERENCES Users(UserID) ON DELETE CASCADE
);
GO

-- 3. Participants Table
CREATE TABLE Participants (
    ParticipantID INT IDENTITY(1,1) PRIMARY KEY,
    UserID INT UNIQUE NOT NULL,
    DateOfBirth DATE NOT NULL,
    ProfilePictureURL NVARCHAR(500) NULL,
    FOREIGN KEY (UserID) REFERENCES Users(UserID) ON DELETE CASCADE
);
GO

-- 4. Events Table
CREATE TABLE Events (
    EventID INT IDENTITY(1,1) PRIMARY KEY,
    OrganiserID INT NOT NULL,
    Name NVARCHAR(100) NOT NULL,
    Description NVARCHAR(500) NOT NULL,
    EventDate DATETIME NOT NULL,
    Location NVARCHAR(200) NOT NULL,
    Distance DECIMAL(5,2) NOT NULL,
    EventType NVARCHAR(20) NOT NULL CHECK (EventType IN ('Run', 'Walk', 'Cycle')),

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        BannerImageUrl NVARCHAR(500) NULL,
        CreatedAt DATETIME DEFAULT GETDATE(),
        FOREIGN KEY (OrganiserID) REFERENCES Organisers(OrganiserID)
    );
GO

-- 5. Categories Table
CREATE TABLE Categories (
    CategoryID INT IDENTITY(1,1) PRIMARY KEY,
    EventID INT NOT NULL,
    Name NVARCHAR(50) NOT NULL,
    Description NVARCHAR(200) NULL,
    MinAge INT NULL,
    MaxAge INT NULL,
    FOREIGN KEY (EventID) REFERENCES Events(EventID) ON DELETE CASCADE
);
GO

-- 6. Enrolments Table
CREATE TABLE Enrolments (
    EnrolmentID INT IDENTITY(1,1) PRIMARY KEY,
    ParticipantID INT NOT NULL,
    EventID INT NOT NULL,
    CategoryID INT NOT NULL,
    RegistrationDate DATETIME DEFAULT GETDATE(),
    Status NVARCHAR(20) DEFAULT 'Pending' CHECK (Status IN ('Pending', 'Confirmed', 'Cancelled')),
    CONSTRAINT UQ_Enrolment UNIQUE (ParticipantID, EventID),
    FOREIGN KEY (ParticipantID) REFERENCES Participants(ParticipantID),
    FOREIGN KEY (EventID) REFERENCES Events(EventID),
    FOREIGN KEY (CategoryID) REFERENCES Categories(CategoryID)
);
GO

-- 7. Results Table
CREATE TABLE Results (
    ResultID INT IDENTITY(1,1) PRIMARY KEY,
    EnrolmentID INT UNIQUE NOT NULL,
    FinishTime TIME NOT NULL,
    FinishPosition INT NOT NULL,
    Status NVARCHAR(20) DEFAULT 'Published' CHECK (Status IN ('Published', 'Pending', 'Disqualified')),
    FOREIGN KEY (EnrolmentID) REFERENCES Enrolments(EnrolmentID)
);
GO

```

3. Database Seed Data (SQL Script)

```

-- SEED DATA

-- Seed Users
INSERT INTO Users (Email, PasswordHash, FullName, Role) VALUES
('organiser1@raceday.com', 'hashed_password_123', 'John Organiser', 'Organiser'),
('organiser2@raceday.com', 'hashed_password_456', 'Sarah Events', 'Organiser'),
('participant1@raceday.com', 'hashed_password_789', 'Tom Runner', 'Participant'),
('participant2@raceday.com', 'hashed_password_abc', 'Lisa Walker', 'Participant'),
('participant3@raceday.com', 'hashed_password_def', 'Mike Cyclist', 'Participant');

```

GO

-- Seed Organisers

INSERT INTO Organisers (UserID, OrganisationName, ContactNumber) VALUES

(1, 'Cape Town Events', '+27 82 123 4567'),

(2, 'Durban Sports', '+27 83 987 6543');

GO

-- Seed Participants

INSERT INTO Participants (UserID, DateOfBirth, ProfilePictureURL) VALUES

(3, '1990-05-15', NULL),

(4, '1985-10-20', NULL),

(5, '1995-03-08', NULL);

GO

-- Seed Events

INSERT INTO Events (OrganiserID, Name, Description, EventDate, Location, Distance, EventType) VALUES

(1, 'Cape Town Cycle Tour', 'Annual cycling event through Cape Town', '2026-03-15 07:00:00', 'Cape Town, Western Cape', 109.00, 'Cycle'),

(1, 'Soweto Marathon', 'Iconic marathon through Soweto', '2026-04-20 06:30:00', 'Soweto, Gauteng', 42.20, 'Run'),

(2, 'Durban Walkathon', 'Scenic coastal walk for charity', '2026-05-10 08:00:00', 'Durban, KwaZulu-Natal', 21.10, 'Walk');

GO

-- Seed Categories

INSERT INTO Categories (EventID, Name, Description, MinAge, MaxAge) VALUES

(1, 'Under 20', 'Cyclists under 20 years old', 15, 19),

(1, 'Senior', 'Cyclists 20-39 years old', 20, 39),

(1, 'Veteran', 'Cyclists 40+ years old', 40, NULL),

(2, 'Under 20', 'Runners under 20', 15, 19),

(2, 'Senior', 'Runners 20-39 years old', 20, 39),

(2, 'Veteran', 'Runners 40+ years old', 40, NULL),

(3, 'Under 20', 'Walkers under 20', 15, 19),

(3, 'Senior', 'Walkers 20-39 years old', 20, 39),

(3, 'Veteran', 'Walkers 40+ years old', 40, NULL);

GO

-- Seed Enrolments

INSERT INTO Enrolments (ParticipantID, EventID, CategoryID, Status) VALUES

(1, 1, 2, 'Confirmed'),

(1, 2, 5, 'Pending'),

(2, 1, 3, 'Confirmed'),

(2, 3, 8, 'Confirmed'),

(3, 1, 1, 'Pending'),

(3, 3, 9, 'Confirmed');

GO

-- Seed Results

INSERT INTO Results (EnrolmentID, FinishTime, FinishPosition) VALUES

(1, '03:45:30', 47),

(3, '04:12:15', 78),

(4, '02:30:45', 12),

(6, '03:55:20', 34);

GO